Pablo Watfi

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ABOUT ME

Machine Learning & Software Engineer with 8+ years of experience designing and deploying scalable ML and data systems across retail, enterprise AI platforms, adtech, marketing, and public sector domains. Proficient in building end-to-end pipelines, including RAG architectures, predictive modeling systems, and optimization frameworks, using Python, Databricks, Airflow, and modern MLOps practices. Project experience includes developing a GenAI-powered RAG system that increased product content generation throughput by 6x for a global sportswear manufacturer, and building a production XGBoost pipeline with MLflow monitoring to optimize campaign targeting and improve model performance and iteration speed.

SKILLS

Programming Languages: Python, Scala, TypeScript, SQL
Databases & Storage: AWS S3, PostgreSQL, Presto
Data Tools: Databricks, Airflow, Pandas, Spark
Programming Languages: Python, Scala, TypeScript, SQL
Visualization Tools: MetaPost, Matplotlib, Seaborn
Cloud: AWS
ML Frameworks & Tools: scikit-learn, HDBSCAN, spaCy, Keras, TensorFlow, XGBoost, SentenceTransformers, MLflow, Databricks Mosaic AI Vector Search, OpenSearch, PuLP
Others: Google Drive API, Kubernetes, Docker

EXPERIENCE

Sr Machine Learning Engineer — *US (remote from Argentina)*
Nike (acquired Datalogue) — *2021-present*

- Built and enhanced a GenAI-powered product copy system using SentenceTransformers (embeddings) and Anthropic LLM (generation), increasing content production from 25 to 160 per week (6x).
- Designed and implemented a Retrieval-Augmented Generation (RAG) system using Databricks Mosaic AI Vector Search, building end-to-end ingestion (document parsing, embedding, indexing) and retrieval pipelines to vectorize product data from presentations; enabled marketers to query specific attributes and retrieve relevant product codes and descriptions, and led trade-off analysis vs. OpenSearch, selecting Mosaic for scalability and native integration.
- Developed, maintained, and optimized an XGBoost-based training and inference pipeline, orchestrated with Airflow, to predict redemption rates for SNKRS app campaigns, and implemented MLflow for experiment tracking and model monitoring, improving campaign effectiveness and iteration speed.
- Influenced system architecture decisions by demonstrating when structured database lookups outperformed vector search for specific product code queries, reducing latency, compute overhead, and infrastructure costs.
- Built an optimization engine using PuLP (Python Linear Programming library) to select target users for SNKRS app campaigns based on predicted purchase intent, improving allocation of limited offers and campaign effectiveness.

Machine Learning Engineer — *Canada/US (remote from Argentina)*
Datalogue — *2018-2021*

- Enhanced and maintained production deep learning models (TensorFlow, Keras) and redesigned training interfaces to support non-ML users, simplifying parameter tuning and guiding data selection, which enabled clients to independently train models while maintaining high performance.
- Developed and maintained scalable back-end microservices using Python and Scala, containerized with Docker and running on Kubernetes, to support core data platform workflows; ramped up in Scala from scratch to contribute to production systems and ensure reliable, high-performance data processing.

- Contributed to front-end development using TypeScript to improve internal tooling and support ML workflows, and maintained a Python SDK to provide an intuitive interface for interacting with platform services.
- Designed and implemented a custom Named Entity Recognition (NER) pipeline using spaCy to automatically detect and obfuscate PII (e.g., names, emails, IDs) in unstructured documents for a global aerospace manufacturer (airliners and helicopters), improving data privacy compliance and reducing manual redaction efforts by 80%.

Data Scientist *Buenos Aires*
Jampp *2017-2018*

- Designed and implemented a user segmentation algorithm using HDBSCAN (scikit-learn), leveraging behavioral and metadata features to improve bid alignment in the advertising platform, enabling more precise targeting and campaign performance optimization.
- Designed and implemented an A/B testing framework to measure the causal impact of advertising campaigns in mobile ad networks, enabling data-driven optimization of campaign performance (e.g., lift in CTR/CVR).
- Built and maintained scalable ETL pipelines using Python and Presto (SQL) to ingest and process data from heterogeneous sources (relational DBs, NoSQL stores, web services, flat files), structuring datasets for flexible and efficient analysis.
- Defined key product and marketing metrics (e.g., CTR, CVR, retention) and developed data visualization dashboards using Matplotlib and Seaborn to monitor performance trends and support decision-making across teams.
- Developed a Python package integrating with the Google Drive API to store and track model outputs and performance metrics, enabling systematic monitoring of model improvements over time.

Research Assistant *Buenos Aires*
Torcuato Di Tella University *2012-2013*

- Developed a profit-maximizing optimization model using AMPL (A Mathematical Programming Language) and lp_solve for one of the largest commodity trading firms in Argentina, achieving 4.5% profit improvement through more efficient trading and logistics decisions.
- Formulated and solved a constrained optimization problem to determine optimal trading, storage, and shipping policies under capacity, network, and transportation cost constraints, enabling better resource allocation across the supply chain.
- Built data ingestion and analysis pipelines to support the optimization model and developed visualizations using MetaPost to communicate results and policy insights to stakeholders.

EDUCATION

Cornell Tech, NYC, United States *2017*
Master in Operations Research and Information Engineering

University of Buenos Aires, Argentina *2015*
Certificate in Data Knowledge & Discovery

University of Buenos Aires, Argentina *2014*
Bachelor in Applied Mathematics

PUBLICATION

2016: N. Merener, R. Moyano, N.E. Stier-Moses, P. Watfi. Optimal Trading and Shipping of Agricultural Commodities [PDF]. Journal of the Operational Research Society, 67:1, 114–126, 2016.

PROJECT

Assistant Professor, Universidad de Buenos Aires - 2002-2006: Taught Calculus and Linear Algebra using lectures, guided problem-solving sessions, and assessments to strengthen analytical thinking and mathematical foundations for undergraduate students.